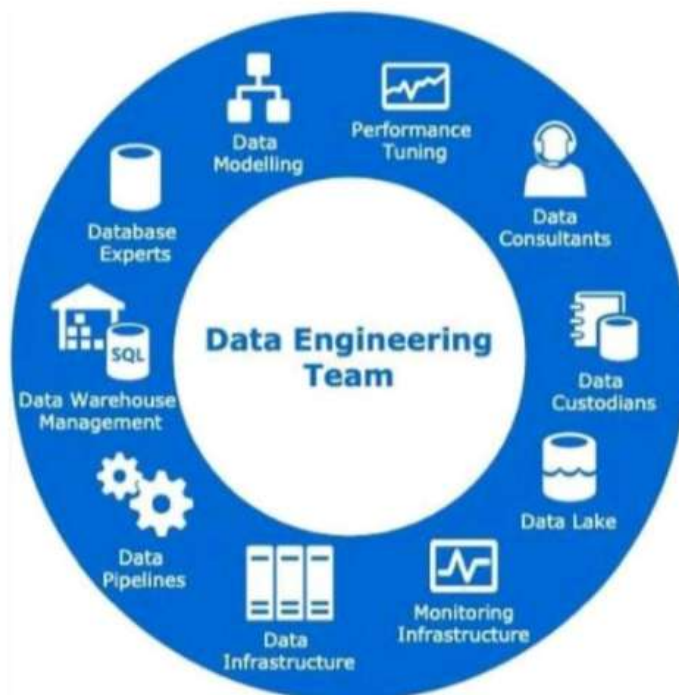


12 DATA ENGINEERING CONCEPTS — INTERVIEW MASTER GUIDE (2026)



1. Data Modeling (OLTP vs OLAP)

What interviewers evaluate

- Understanding of workload patterns
- Ability to design for performance and scale

Interview answer

OLTP systems are optimized for frequent inserts, updates, and deletes using normalized schemas, whereas OLAP systems are optimized for analytical queries using denormalized schemas such as star or snowflake models.

Follow-up questions

- Why is OLTP unsuitable for analytics?
Too many joins and poor query performance for aggregations.
- What is a star schema?
A central fact table connected to multiple dimension tables.

2. ETL vs ELT

What interviewers test

- Cloud data warehouse knowledge
- Cost and scalability awareness

Interview answer

In modern cloud platforms, ELT is preferred because raw data is first loaded into the warehouse and transformations leverage scalable compute.

Follow-ups

- When is ETL still preferred?
Complex transformations, compliance rules, or legacy systems.
- Examples of ELT tools
Snowflake, BigQuery, Redshift.

3. Data Warehouse Design (Fact & Dimension)

Interview expectation

Business understanding over definitions.

Interview answer

Fact tables store quantitative metrics at a defined grain, while dimension tables store descriptive attributes for slicing and filtering data.

Follow-ups

- What is grain?
The lowest level of detail stored in a fact table.
- Why is grain important?
Incorrect grain leads to data duplication and incorrect aggregates.

4. Slowly Changing Dimensions (SCD)

Frequently asked concept

Handling historical data changes.

Types

- Type 1: Overwrite data, no history
- Type 2: Maintain full history using surrogate keys and dates
- Type 3: Limited history

Interview answer

For tracking historical changes such as customer address, SCD Type 2 is used with effective start and end dates.

5. SQL Window Functions

Interview focus

Advanced SQL and analytical thinking.

Must-know functions

ROW_NUMBER, RANK, DENSE_RANK, LAG, LEAD, running aggregates.

Interview answer

Window functions perform calculations across related rows without collapsing the result set.

Common use cases

Deduplication, ranking, trend analysis, latest record selection.

6. Batch Processing vs Streaming

Interview goal

Understanding latency and cost trade-offs.

Interview answer

Batch processing handles large volumes of data at scheduled intervals, while streaming processes data in near real time for low-latency use cases.

Follow-ups

- Can both coexist?
Yes, using Lambda or Kappa architectures.
- Why not use streaming everywhere?
Higher cost and operational complexity.

7. Apache Spark Fundamentals

Interviewer focus

Performance and optimization knowledge.

Key concepts

Lazy evaluation, transformations vs actions, shuffles, partitioning, caching.

Common trap

Saying Spark is always faster than SQL.

Correct answer: Spark efficiently executes SQL at scale when optimized.

8. Data Lake vs Data Warehouse vs Lakehouse

Interview answer

A data lake stores raw data in its native format, a data warehouse stores structured, curated data for analytics, and a lakehouse combines both using open formats.

Follow-ups

- Why Parquet over CSV?
Columnar storage, compression, faster queries.
- What is schema-on-read?
Schema applied at query time.

9. Partitioning and Indexing

Interview focus

Query performance and cost optimization.

Interview answer

Partitioning reduces the amount of data scanned, while indexing speeds up point lookups.

Follow-ups

- Best partition column
Date.
- Poor partition choice
High-cardinality columns like user_id.

10. Data Quality and Validation

Interviewer expectation

Production-level responsibility.

Common checks

Null validation, duplicates, schema drift, referential integrity.

Interview answer

Data quality checks should fail the pipeline early rather than allowing bad data to reach dashboards.

11. Orchestration and Scheduling

Interview focus

Reliability and failure handling.

Must-know concepts

DAGs, retries, backfills, idempotency.

Interview answer

Idempotent pipelines ensure reruns do not create duplicate data.

12. Security and Cost Optimization

Senior-level topic

Especially important for 3+ years of experience.

Must-cover areas

IAM and RBAC, encryption at rest and in transit, secrets management, cost monitoring.

Interview answer

Security and cost optimization are continuous responsibilities and must be built into pipeline design.